

# Measuring the Mechanical Index and Acoustic Intensities of a Custom Verasonics Sequence

Hydrophone measurement protocol

TU Eindhoven — Shear-wave elastography

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## Abstract

Any custom transmit sequence used on a subject must be shown to stay inside the accepted acoustic-output limits. The Verasonics software reports these for a handful of its own example scripts, but not for sequences we write ourselves, and not for the long ARF pushes used in shear-wave elastography — so they have to be measured. This document is a protocol for doing that with a calibrated hydrophone: what the setup is (§3), how to capture a voltage correctly (§4), how to convert it to MI,  $I_{\text{sppa.3}}$  and  $I_{\text{spta.3}}$  (§5), and the step-by-step measurement itself (§6); Figure 5 is the whole thing on one page. §2 sets out both the *ideal* measurement — a full 3-D raster of the field — and the reduced one-dimensional measurement that is actually described here, and what it costs. §7 collects the mistakes that are easy to make and expensive to discover late. §8 gives worked results from our own L11-5 and S5-1 measurements as examples of what the output looks like. Appendix A covers the motion stage the hydrophone is mounted on, and Appendix B a separate comparison of our two S5-1 probes.

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## Symbols and abbreviations

The equations in §1 and §5 use these throughout. A subscript .3 on any acoustic quantity means it has been **derated** by  $0.3 \text{ dB cm}^{-1} \text{ MHz}^{-1}$  to emulate the attenuation of intervening tissue; without it the quantity is the one measured in water.

### The regulated quantities

|                     |                                                                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MI                  | Mechanical index, $p_{r,3}/\sqrt{f_{\text{awf}}}$ . A <i>cavitation</i> index: the potential for mechanical damage. Dimensionless.                                      |
| $I_{\text{sppa},3}$ | Spatial-peak <i>pulse-average</i> intensity, $\text{PII}_{,3}/\text{PD}$ [ $\text{W cm}^{-2}$ ]. A <i>thermal</i> quantity: how hard one pulse heats.                   |
| $I_{\text{spta},3}$ | Spatial-peak <i>time-average</i> intensity, $\text{PII}_{,3} \cdot \text{PRF}$ [ $\text{mW cm}^{-2}$ ]. The long-time average, so it depends on how often you transmit. |
| TI                  | Thermal index. Not measured here; mentioned because $I_{\text{spta}}$ does not capture the temperature rise <i>within</i> a burst (§6.6).                               |

### What goes into them

|       |                                                                                                                                                     |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| $p_r$ | Peak rarefactional pressure [MPa]: the most negative excursion of the pressure waveform. <i>Not</i> $V_{\text{pp}}/2$ — the waveform is asymmetric. |
| $p_+$ | Peak compressional pressure: the most positive excursion. Reported for context; no index uses it.                                                   |

|                  |                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| $f_{\text{awf}}$ | Acoustic working frequency [MHz]: the centre of the $-6$ dB band of the fundamental (§5.3). Measure it, do not assume the programmed transmit frequency. |
| PII              | Pulse intensity integral [ $\text{J m}^{-2}$ ], $\int p^2/(\rho c) dt$ over the pulse.                                                                   |
| PD               | Pulse duration [s], $1.25(t_{90} - t_{10})$ .                                                                                                            |
| $t_{10}, t_{90}$ | The instants at which the cumulative intensity integral reaches 10 % and 90 % of its final value.                                                        |
| PRF              | Pulse repetition frequency [Hz]. A property of the <i>sequence</i> , not of a capture. For a burst protocol use the <i>effective</i> rate.               |
| $\rho, c$        | Density and speed of sound of the medium; $1000 \text{ kg m}^{-3}$ and $1500 \text{ m s}^{-1}$ throughout.                                               |
| $z$              | Depth [cm]: distance from the transducer face to the measurement point, $\ \text{Pos} - \text{Home}\ $ (§3.4). Sets the derating.                        |

### The hydrophone chain (§5.1)

|                                  |                                                                                                                                                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| $M_c(f)$                         | <b>Open-circuit</b> end-of-cable hydrophone sensitivity [ $\text{V Pa}^{-1}$ ] — what the Onda certificate gives. Not what the scope sees.           |
| $M_L(f)$                         | <b>Loaded</b> system sensitivity [ $\text{V MPa}^{-1}$ ] — $M_c$ after the capacitive divider and any gain. This is what converts volts to pressure. |
| $G(f)$                           | Gain of the AH-2010 pre-amplifier ( $\approx 20$ dB), referenced to a $50 \Omega$ load.                                                              |
| $C_H$                            | Hydrophone element capacitance, $\approx 12.7$ pF.                                                                                                   |
| $C_A$                            | Pre-amplifier input capacitance, 6.1 pF. Loads the hydrophone on chain A.                                                                            |
| $C_{\text{load}}$                | Cable + scope capacitance, $\approx 123$ pF. Loads the hydrophone on chain B, where it dominates. Not a datasheet value — measure it (§5.2).         |
| $R_L$                            | Oscilloscope input impedance, $50 \Omega$ or $1 \text{ M}\Omega$ (§4.3).                                                                             |
| $R_{\text{out}}, R_{\text{ref}}$ | Pre-amplifier output impedance and the load its calibration is referenced to; both $50 \Omega$ .                                                     |
| $V_A, V_B$                       | The same acoustic point read on chain A and chain B.                                                                                                 |
| $V_{\text{pp}}$                  | Peak-to-peak voltage of a capture.                                                                                                                   |
| Chain A                          | Hydrophone $\rightarrow$ pre-amplifier $\rightarrow$ scope. Sensitive; clips at $\approx 2.2$ V (§4.4).                                              |
| Chain B                          | Hydrophone $\rightarrow$ scope, pre-amplifier removed. $73\times$ less sensitive; no ceiling (§4.2).                                                 |

### Beam geometry (§6.4)

|           |                                                                                                                                                        |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\lambda$ | Acoustic wavelength at $f_{\text{awf}}$ .                                                                                                              |
| $F_{\#}$  | f-number of the transmit beam, focal depth divided by active aperture.                                                                                 |
| FWHM      | Full width at half maximum of the beam, $\approx 1.21 \lambda F_{\#}$ in the focal plane. The transverse search step is expressed as a fraction of it. |
| $\Delta$  | Transverse scan step.                                                                                                                                  |
| Home, Pos | Stage coordinates of the transducer face centre and of the hydrophone tip, written into every capture note.                                            |

### Abbreviations

|      |                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------|
| ARF  | Acoustic radiation force — the long, high-amplitude push that drives a shear wave.                                                       |
| FDA  | US Food and Drug Administration; the source of the output limits used here.                                                              |
| IEC  | International Electrotechnical Commission; defines $f_{\text{awf}}$ and the derating.                                                    |
| .hws | Hierarchical Waveform Storage, the NI capture format. HDF5 underneath.                                                                   |
| SFP  | Soft Front Panel, the NI application that drives the digitizer and writes .hws.                                                          |
| ADC  | Analogue-to-digital converter. The PCI-5112 is <b>8-bit</b> , which is why the vertical range must be set tightly.                       |
| RIS  | Random interleaved sampling; equivalent-time sampling, valid only for repetitive signals and <b>never</b> for a single-shot push (§4.6). |

# 1 What has to be measured

Three quantities are regulated. All are defined at the *spatial peak* of the field, and all are *derated* by  $0.3 \text{ dB cm}^{-1} \text{ MHz}^{-1}$  to emulate the attenuation of intervening tissue (the subscript .3):

$$\text{MI} = \frac{p_{r.3}}{\sqrt{f_{\text{awf}}}} \quad \text{peak rarefactional pressure [MPa] over } \sqrt{f} \text{ [MHz]}, \quad (1)$$

$$I_{\text{sppa.3}} = \frac{\text{PII}_{.3}}{\text{PD}} \quad \text{pulse-average intensity,} \quad (2)$$

$$I_{\text{spta.3}} = \text{PII}_{.3} \cdot \text{PRF} \quad \text{time-average intensity.} \quad (3)$$

MI is a **cavitation** (mechanical) index; the two intensities are **thermal**. All symbols are defined in §5.

## 1.1 The limits

The FDA guidance<sup>1</sup> sets application-specific limits (Table 2). Pick the row for the intended use and compare against it.

**Table 2:** FDA acoustic output limits by application. Note the **or** in the last two columns: formally a device needs  $I_{\text{spta.3}}$  below its limit *and* either MI or  $I_{\text{sppa.3}}$  below theirs. Satisfying *both* is the conservative choice and is what is assumed throughout this document.

| Use                   | $I_{\text{spta.3}}$ ( $\text{mW cm}^{-2}$ ) | $I_{\text{sppa.3}}$ ( $\text{W cm}^{-2}$ ) | <b>or</b> MI |
|-----------------------|---------------------------------------------|--------------------------------------------|--------------|
| Peripheral vessel     | 720                                         | 190                                        | 1.9          |
| Cardiac               | 430                                         | 190                                        | 1.9          |
| Fetal imaging & other | 94                                          | 190                                        | 1.9          |
| Ophthalmic            | 17                                          | 28                                         | 0.23         |

**Cardiac** covers transthoracic adult and paediatric use; **peripheral vessel** includes the vessels of the neck; **fetal imaging & other** is the catch-all (abdominal, small organ, musculoskeletal, neonatal and adult cephalic). Transcranial use above  $94 \text{ mW cm}^{-2}$  additionally requires an estimate of the maximum temperature rise from a model that accounts for skull heating.

One caveat on what these numbers are for: the FDA compliance policy explicitly **excludes shear-wave elastography**, and assumes 1–100 cycles per pulse — a 1500–1900-cycle ARF push is an order of magnitude outside that. For an ARF sequence, Table 2 is therefore a **benchmark rather than a compliance route**: these are the right quantities to measure and the right targets to stay under, but the approval route is the medical-ethics committee.

### Common mistake — quoting “the” limit without saying which application

$I_{\text{spta.3}}$  ranges over  $40\times$  across the rows of Table 2 —  $720 \text{ mW cm}^{-2}$  for peripheral vessel against  $17 \text{ mW cm}^{-2}$  for ophthalmic. State the application alongside every number you report, and when it is not settled which applies, report against the stricter one.

## 1.2 What changes what

This table is worth internalising before planning a measurement, because it determines what must be measured and what can be scaled:

<sup>1</sup>Marketing Clearance of Diagnostic Ultrasound Systems and Transducers, FDA, 2023; a copy is in docs/datasheets/.

| Change                         | MI                                            | $I_{\text{sppa.3}}$   | $I_{\text{spta.3}}$   |
|--------------------------------|-----------------------------------------------|-----------------------|-----------------------|
| Transmit voltage $V$           | $\propto V$                                   | $\propto V^2$         | $\propto V^2$         |
| Pulse length (# cycles)        | unchanged                                     | unchanged             | $\propto$ cycles      |
| PRF / duty cycle               | unchanged                                     | unchanged             | $\propto$ PRF         |
| Aperture (# elements), focused | $\approx \propto N$                           | $\approx \propto N^2$ | $\approx \propto N^2$ |
| Beam type, focal depth         | changes the field shape — must be re-measured |                       |                       |

The pulse-length row is why a long ARF push is “free” on MI, and why long pushes tend to be limited by  $I_{\text{sppa.3}}$  rather than by MI. The first three rows are what allow a single measurement at the field maximum to be extrapolated instead of re-measuring the whole field for every variant.

## 2 Two ways to do this

### 2.1 The ideal measurement: a full 3-D raster

What one really wants is the acoustic field *sampled on a 3-D grid*, with several waveforms saved at every grid point. Every quantity then follows without assumption: the spatial peak is simply the maximum over the volume, the beam width and cross-sectional area come out for free (needed for the spatial-averaging correction and for the thermal indices), the repeatability at each point is measured rather than estimated, and no search algorithm has to converge on anything.

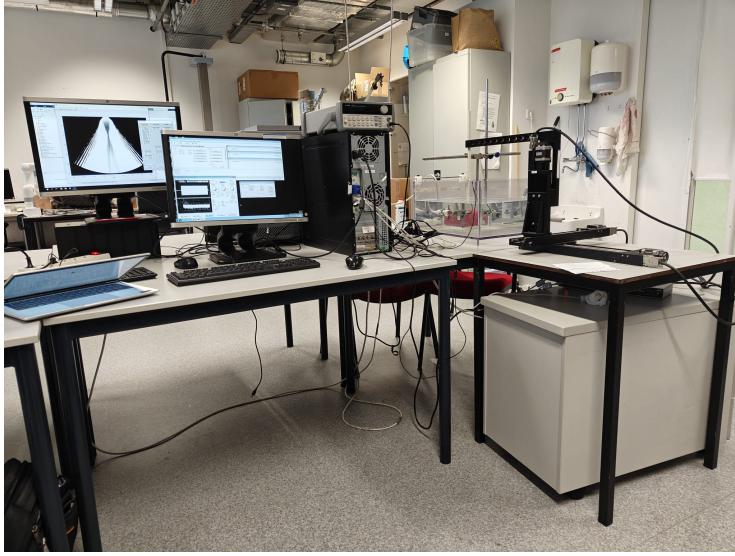
The hardware for this already exists in the setup. **OWIsoft supports a meander function** that drives the stage through a predefined grid (Appendix A), so the motion side is solved. **The only real obstacle is synchronisation:** the Verasonics, the stage and the digitizer must be coordinated so that, at every grid point, the stage settles, the sequence transmits, and the scope captures and stores several records under a filename carrying the coordinates. Sketching what that needs:

- a **position-reached signal** from the stage (a TTL output, or a polled status flag over the OWIsoft interface) to gate acquisition, so that no capture is taken while the stage is moving or still ringing down;
- **programmatic scope control** instead of the Soft Front Panel — arm, capture  $N$  records, write to a file named by grid index — driven from the same script that steps the meander;
- a **dwel time** per point long enough for the stage to settle and for  $N$  transmit events to occur, which sets the total scan duration: a  $20 \times 20 \times 20$  grid at 2 s per point is over four hours, so the grid extent and step have to be chosen with that in mind;
- a **coordinate log** written by the same script, so that positions are never inferred afterwards.

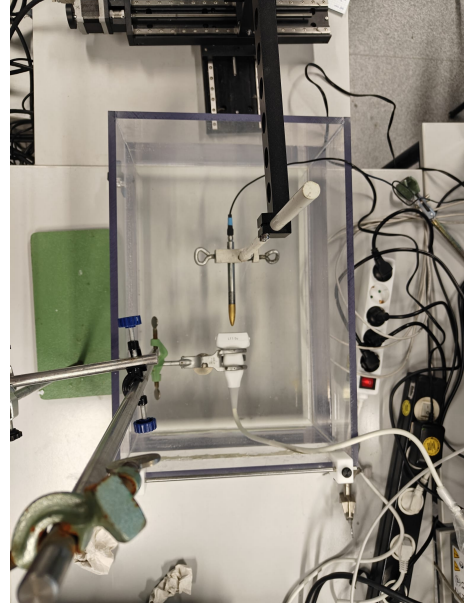
**This has not been built.** It is a significant piece of integration work and there was no time for it. It is the right thing to build if these measurements are going to be repeated often, and this section exists so that whoever picks it up knows the pieces are all present.

### 2.2 The practical measurement: an axial line with a local transverse search

In the absence of that synchronisation, the measurement described in this document reduces the 3-D raster to a line: step along the beam axis, and at each axial station move laterally and in elevation by hand (or by short stage moves) to find the local maximum, saving only that peak waveform. This is a poor-man’s version of the ideal scan and it should be understood as such.



(a) The bench. Left screen: the Verasonics running the measurement sequence. Middle screen: the digitizer's Soft Front Panel and the stage control. The digitizer is a card inside the PC tower; the water tank and the 3-axis stage are on the bench to the right.



(b) The tank. The probe is held in a simple clamp on a laboratory stand; the hydrophone hangs from the arm of the 3-axis stage, tip facing the probe. Nothing else is in the water path.

**Figure 1:** The measurement setup.

|                                                      | 3-D raster (ideal)                           | Axial line + search (this document)                            |
|------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------|
| Spatial peak                                         | found, by construction                       | found only if the transverse search converges at every station |
| Beam area / width                                    | measured                                     | not measured; spatial-averaging correction must be modelled    |
| Repeatability                                        | measured per point                           | not measured unless repeats are taken deliberately             |
| Off-axis structure (grating lobes, secondary maxima) | visible                                      | <b>invisible</b>                                               |
| Effort                                               | hours of scanning, plus the integration work | an afternoon                                                   |

The one substantive risk in the reduced measurement is the last row: a secondary maximum somewhere off the searched line would simply not be seen. For a centred, unsteered beam from a well-behaved array this is unlikely to exceed the main lobe, which is what makes the reduction defensible — but it is an assumption, not a result.

## 3 Equipment and setup

### 3.1 The measurement chain

#### Hydrophone — Onda HGL-0400, SN 1037.

Capsule hydrophone with a  $400\ \mu\text{m}$  nominal sensitive element. The small element is what makes it usable at a tight focus: it averages the pressure over its aperture, so a larger element under-reads the spatial peak. Open-circuit sensitivity  $M_c \approx 200\ \text{nV Pa}^{-1}$  ( $\approx -254\ \text{dB re } 1\ \text{V } \mu\text{Pa}^{-1}$ )

and element capacitance  $C_H \approx 12.7\text{ pF}$ , both weakly frequency dependent (Figure 3). Calibrated by Onda on 10-Jun-2024; certificates in `calibration/`.

#### Pre-amplifier — Onda AH-2010-025, SN 1109.

Nominal 20 dB gain, 25 MHz bandwidth, input capacitance  $C_A = 6.1\text{ pF}$ , **output impedance  $50\ \Omega$** . Two properties of this box dominate the practical side of the measurement: its gain is specified *into a  $50\ \Omega$  load* (§4.3), and its output **clips at 4 V peak-to-peak into  $50\ \Omega$**  (§4.4).

#### Digitizer — National Instruments PCI-5112.

8-bit, 100 MHz analogue bandwidth,  $100\text{ MS s}^{-1}$  real-time, 16 MB to 32 MB per channel; a card in the PC, operated through the NI *Scope Soft Front Panel* (Scope-SFP 3.8), which writes `.hws` files. Its input impedance is **software-selectable** between  $50\ \Omega$  and  $1\text{ M}\Omega$ , which makes the impedance cross-check of §4.3 a keystroke rather than a cable change.

Being **8-bit** is its one real limitation: only  $\approx 256$  codes across the vertical range, so the range must be set tightly around the signal. Quantisation shows up in the data as identical readings at neighbouring positions on a flat part of the field.

#### Positioning — 3-axis motorised translation stage, controlled by OWIsoft.

The stage carries the hydrophone; the probe is fixed. All coordinates in this document are stage coordinates in mm. OWIsoft's meander function — which drives a predefined grid — is the basis of the ideal measurement of §2.1.

#### Tank and mounting.

Nothing special is required of the tank: a clear acrylic box large enough to hold the probe, the intended stand-off distance and the hydrophone, with room to move the hydrophone past the expected field maximum in all three axes. Probe and hydrophone are each held in a simple clamp — the probe on a laboratory stand, the hydrophone on the stage arm (Figure 1b).

#### Function generator (optional).

Useful for an end-to-end electrical check of the acquisition chain: feed a known sine into the pre-amplifier input and verify the recorded amplitude and the  $50\ \Omega$  /  $1\text{ M}\Omega$  behaviour before trusting any acoustic number.

### 3.2 Water

Ordinary tap water is adequate. Ideally it is degassed and, in any case, left to stand so that dissolved gas and suspended bubbles settle out — bubbles on the hydrophone tip or the probe lens attenuate erratically and cause a low reading that is easily mistaken for a positioning error. There is no strict protocol; filling the tank the day before and wiping the tip and the lens before starting covers most of it. Before a session, check both surfaces visually.

### 3.3 Reflections

The tank walls are strong reflectors — acrylic against water returns roughly 40 % of the incident amplitude, the free surface close to 100 %. Whether that matters depends on the **pulse length relative to the round-trip time** to the nearest surface. A short imaging pulse of a few microseconds arrives long before its own reflection and can simply be gated. A long ARF push cannot: an  $850\ \mu\text{s}$  burst occupies over a metre of water path, so in any bench-scale tank the reflection overlaps the direct field, and the resulting error is position-dependent — it distorts the shape of a measured profile, not just its scale.

**Line the tank boundaries in the beam path with an acoustic absorber** (rubber tiles, or open-cell foam as a poorer substitute) whenever long bursts are measured — at minimum the wall opposite the probe and the bottom. Whether it is strictly needed in a given tank is testable from the waveforms already saved: a reflection **steps** the burst envelope when it arrives and leaves a **tail** after the transmit stops, and neither is present if the tank is clean. `MakeReportFigures`

implements that check. Our own untreated tank turned out to be clean to better than 1 %, but that result is specific to one tank, probe and geometry and is not a general licence to skip the absorber.

### 3.4 Geometry and the Home convention

The hydrophone is carried by a three-axis OWIS stage; Appendix A is how to drive it, including where Home is set. Define, at the start of every session,

Home = the stage coordinate at which the hydrophone tip sits at the **centre of the transducer face**,

and store both Home and the current position Pos in every capture, so that the derating depth is

$$z = \|\text{Pos} - \text{Home}\|. \quad (4)$$

Derating is exponential in  $z$  (Eq. 9); at 2 MHz and 7 cm a 5 mm error in the assumed origin is already 0.3 dB on MI and 0.6 dB on the intensities. Fixing the origin at the transducer face centre and recording it in every file removes the ambiguity.

Alignment of the probe to the stage axes need not be perfect, and no effort should be wasted making it so. The procedure of §6 *measures around* the misalignment by searching transversely at every axial station; a small tilt then costs extra search range, not accuracy.

#### Common mistake — using an axis reading as the depth

The hydrophone is generally not exactly on the beam axis, so a single-axis coordinate difference under-states the propagation distance. Use the Euclidean distance of Eq. 4.

## 4 Capturing a voltage

The hydrophone is a capacitive source producing a few hundred  $\mu\text{V}$  to a few mV per MPa. There are two ways to get that onto the digitizer, and a full characterisation generally needs **both**, because they cover different parts of the transmit-voltage range.

### 4.1 Chain A: with the pre-amplifier

The AH-2010 provides 20 dB of gain, and because its input capacitance  $C_A = 6.1 \text{ pF}$  is small compared with the hydrophone's own  $C_H \approx 12.7 \text{ pF}$ , only about a third of the signal is lost in the capacitive divider. The resulting system sensitivity is of order  $1.3 \text{ V MPa}^{-1}$  (Figure 3d). This is the chain to use at low transmit voltage, and the only one usable when the signal is weak.

### 4.2 Chain B: without the pre-amplifier

Above the clipping ceiling, remove the pre-amplifier and connect the hydrophone directly to the high-impedance scope input. There is then no gain and — crucially — the bare hydrophone is loaded by the **entire cable and scope capacitance**  $C_{\text{load}}$  instead of the amplifier's 6.1 pF. Since  $C_{\text{load}}$  is of order 100 pF, an order of magnitude above  $C_H$ , most of the signal is divided away and the sensitivity falls by roughly two orders of magnitude, to tens of  $\text{mV MPa}^{-1}$ .

$C_{\text{load}}$  **is not a datasheet number, and it dominates the result.** It must be determined by measurement — see §5.2, which is a mandatory calibration step before any chain-B result means anything.

### 4.3 Oscilloscope input impedance

The AH-2010 has a  $50\ \Omega$  *output* impedance, and its gain — hence the Onda combined system sensitivity — is referenced to a  $50\ \Omega$  *load*. Terminate that output into a high-impedance ( $1\ \text{M}\Omega$ ) scope input instead and the resistive divider disappears: the scope reads **twice** the voltage the calibration assumes.

The correction, applied to the measured voltage before anything else, is

$$V_{50\ \Omega\text{-equiv}} = V_{\text{meas}} \cdot \frac{R_{\text{ref}}/(R_{\text{ref}} + R_{\text{out}})}{R_L/(R_L + R_{\text{out}})}, \quad R_{\text{ref}} = R_{\text{out}} = 50\ \Omega, \quad (5)$$

i.e.  $\times 0.5$  for  $R_L = 1\ \text{M}\Omega$  and  $\times 1.0$  for  $R_L = 50\ \Omega$ . In `CalculateSafety.m` this is the `ScopeImpedance_Ohm` argument. It applies **only with the pre-amplifier** in the chain; without it there is no  $50\ \Omega$  source and the loading is capacitive instead (§4.2).

**Rules.**

- **Measure into  $50\ \Omega$**  whenever the signal allows it; the calibration then applies directly and there is nothing to correct.
- **Record the impedance setting with every single capture.**
- **Verify the 2 : 1 ratio once per session** on an unsaturated signal by toggling  $50\ \Omega \leftrightarrow 1\ \text{M}\Omega$  without touching anything else. This validates the entire resistive part of the calibration in one keystroke.

#### Common mistake — a factor of two, invisible in the waveform

Forgetting the load correction inflates every pressure and MI by  $2\times$  and every intensity by  $4\times$ , and nothing about the recorded trace looks wrong. Two independent fingerprints reveal the scope was on high impedance: the pre-amplifier clipping ceiling appears at  $\approx 4.4\ \text{V}$  instead of  $\approx 2.2\ \text{V}$ , and the ratio between chains A and B comes out  $2\times$  the expected value. **Trust these over the logbook** — a mis-recorded impedance setting is easy, and the physical evidence is not.

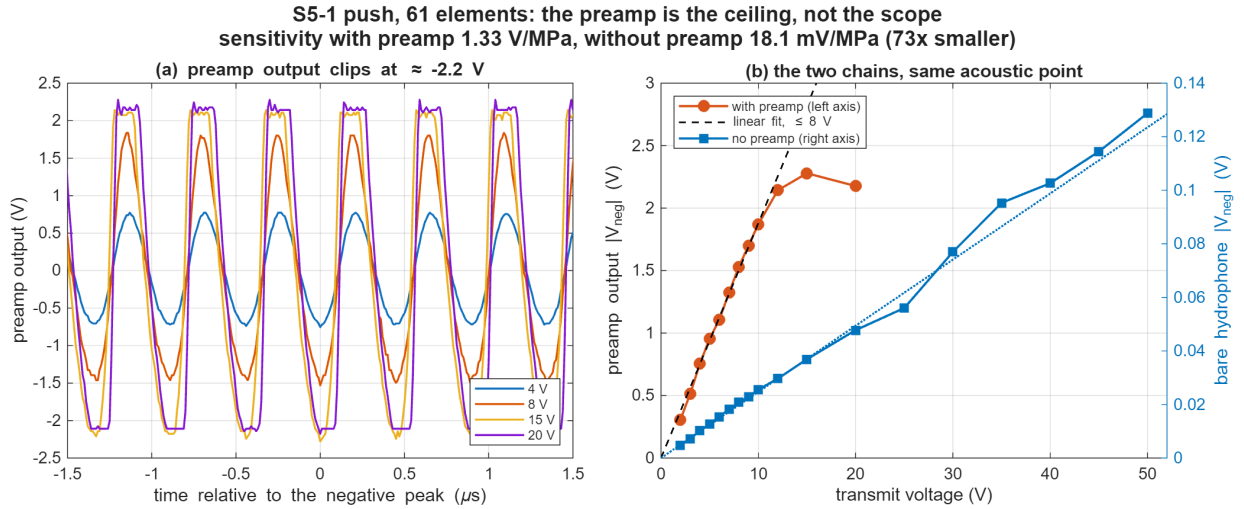
### 4.4 Pre-amplifier clipping

The AH-2010 output saturates at  $4\ \text{V}$  peak-to-peak into  $50\ \Omega$ , i.e.  $\approx 2\ \text{V}$  peak — so a chain-A capture is only meaningful below that. Three consequences:

1. **The pre-amplifier is the ceiling, not the scope.** **An inline attenuator does not help:** it attenuates *after* the stage that is already clipping. Removing the pre-amplifier (chain B) does help.
2. **Either extrapolate or switch chains.** Since MI is linear in transmit voltage, a linear fit through the unsaturated region extrapolates it reliably. Measuring the high-voltage points on chain B is better where the sequence permits it, because it captures nonlinear saturation and transmit-supply sag, which extrapolation cannot.
3. **Overlap the two chains** over several transmit voltages so they can be cross-checked against each other, and so  $C_{\text{load}}$  can be determined (§5.2).

#### Common mistake — judging clipping from the peak value alone

Peak voltage versus drive bends over smoothly rather than stopping dead, so a saturating point can look like an unremarkable data point on a slightly sub-linear curve. **Always look**



**Figure 2:** What clipping looks like, and how the two chains compare. (a) Pre-amplifier output at four transmit voltages, aligned on the negative peak. The lowest trace is a clean (already nonlinearly distorted) acoustic waveform; the upper two are hard-limited at  $\pm 2.2$  V and all amplitude information above that is destroyed. (b) Peak negative voltage versus transmit voltage for both chains at the same acoustic point. Chain A (left axis) departs from its low-voltage linear fit as it saturates; chain B (right axis) stays linear much further, and the mild flattening at its top end is genuine acoustic saturation and transmit-supply sag rather than instrument clipping. Data: S5-1 push, 61 elements.

**at the waveform:** a clipped trace has visibly flat tops and bottoms (Figure 2a). Discard every capture that shows them.

#### 4.5 Save the whole waveform

The scope will display a peak-to-peak number, and it is tempting to record only that. **Do not.** Save the complete digitised record of every capture. Of the quantities in §5, only  $p_r$  can be recovered from a peak reading; the acoustic working frequency, the pulse duration, the pulse-intensity integral, the harmonic content, the reflection check of §3.3 and any later re-analysis with a corrected calibration all need the waveform. Re-measuring is expensive; storage is not.

The Scope-SFP writes `.hws` (*Hierarchical Waveform Storage*), which is **HDF5** underneath and can be read with plain `h5read` — no NI drivers required. Samples are stored as raw 8-bit ADC codes and must be scaled by the stored polynomial  $V = c_0 + c_1 \cdot \text{raw}$ ; `readHWS.m` does this and also decodes the scope configuration and the free-text note. Table 3 is the note to fill in at every capture; it is what lets the analysis reconstruct depth, aperture and drive without a separate logbook.

Only three fields are load-bearing. `Home` and `Pos` give the derating depth  $\|\text{Pos} - \text{Home}\|$ , and `Scope_impedance` selects the load correction of §4.3 — a factor 2 on MI and 4 on the intensities, invisible in the waveform itself. `Pre-amp` is the one field with a *default*: if it is absent the pre-amplifier is taken to have been in the chain, which is the convention the earlier sessions were recorded under. Write it anyway.

**If a required field is missing.** The software does not guess. `CaptureConditions.m` resolves the conditions for every capture and, when the note does not carry one of the three, **asks for it** — a menu for the scope impedance, a number for the depth — and then labels that value in its output as having been supplied rather than recorded. Running non-interactively (`matlab -batch`), where there is nobody to ask, it raises an error naming the missing field and the argument that supplies it instead. Either way the value can also be passed directly (`Depth_cm`, `ScopeImpedance_Ohm`, `WithPreamp`), which is the route to take when re-analysing an old capture whose conditions are

**Table 3:** The note to write into every `.hws` capture. **Required** fields carry information that cannot be recovered from anywhere else and without which the indices cannot be computed; the rest is provenance, and never enters the arithmetic.

| Field           | Example                                           | Required   |
|-----------------|---------------------------------------------------|------------|
| Home            | (55.8, 105.84, -17.3) mm (transducer face centre) | <b>yes</b> |
| Pos             | (46.1, 70.84, -17.3) mm (hydrophone tip)          | <b>yes</b> |
| Scope_impedance | 50 / 1e6                                          | <b>yes</b> |
| Pre-amp         | Yes / No                                          | default    |
| Probe           | S5-1                                              | no         |
| TX_voltage      | 20V                                               | no         |
| Pulse           | normal / inverted                                 | no         |
| PushCycles      | 1900                                              | no         |
| PushElements    | 61                                                | no         |

known from a logbook rather than from the file.

#### Common mistake — answering the impedance prompt from memory

The prompt exists to keep an incomplete capture usable, not to make the field optional. An answer recalled at the end of a long session is exactly the kind of thing that is wrong by a factor of two, and nothing downstream can catch it. **Write the field at acquisition time**, when the setting is in front of you.

## 4.6 Scope settings

- **Real-time sampling at  $100 \text{ MS s}^{-1}$ .** **Never use RIS** (random interleaved sampling): any rate above  $100 \text{ MS s}^{-1}$  on the 5112 is equivalent-time sampling, which reconstructs one waveform from many repetitions. That is valid for a repetitive imaging pulse and **invalid for a single-shot push**. `readHWS.m` warns when it detects the RIS flag.
- **Record length longer than the pulse.** A 1900-cycle push at 2.2 MHz lasts  $850 \mu\text{s}$ ; record  $1000 \mu\text{s}$  ( $10^5$  samples at  $100 \text{ MS s}^{-1}$ ). A truncated record silently under-reports PII and hence both intensities. `readHWS.m` warns if the record is shorter than the expected pulse.
- **Triggering.** Self-trigger on the hydrophone signal (negative slope, level below the smallest expected peak) so the Verasonics can transmit freely with no trigger wire; set a holdoff of about one record length to avoid re-triggering on burst ringing. With an external trigger instead, set the delay to the time of flight  $z/c$  so the window lands on the arrival.
- **Vertical range as tight as the signal allows**, and re-check it after every change of transmit voltage. With 8 bits, a range set  $4\times$  too wide throws away two bits.
- **DC coupling**; subtract the median of the record as the baseline in post-processing.

## 5 From voltage to MI and the intensities

All of this is implemented in `CalculateSafety.m`.

### 5.1 Calibration: the loaded sensitivity

The Onda certificate for the HGL-0400 is an **open-circuit** calibration (`ELECTRICAL_LOAD OpenCircuit`): it gives  $M_c(f)$ , the sensitivity with nothing attached, and the element capacitance  $C_H(f)$ . To use it, the load that is actually attached must be added. With the pre-amplifier that load is its

input capacitance  $C_A(f)$ , giving Onda’s “parallel circuit” loaded sensitivity (Eq. 2a of the Onda *HydroCalMethod* note):

$$M_L(f) = G(f) M_c(f) \frac{C_H(f)}{C_H(f) + C_A(f)}, \quad (6)$$

and without it, the cable-and-scope capacitance takes the place of  $C_A$  and there is no gain:

$$M_L^{\text{no preamp}}(f) = M_c(f) \frac{C_H(f)}{C_H(f) + C_{\text{load}}}. \quad (7)$$

Both  $M_c$  and  $G$  are frequency dependent, and the tabulated certificates (`calibration/`, digested into `HydrophoneTables.mat`) should be evaluated at the *measured*  $f_{\text{awf}}$ , not at a nominal value. Figure 3 shows all of it. Over 1 MHz to 20 MHz the hydrophone sensitivity spans about 4 dB with a sharp dip near 6 MHz, and the preamp gain droops from 19.95 dB to 19 dB; the combined  $M_L$  varies from  $1.05 \text{ V MPa}^{-1}$  to  $1.6 \text{ V MPa}^{-1}$  across that span, though only by a few per cent between neighbouring frequencies in the 2 MHz to 5 MHz range.

#### Common mistake — dropping the capacitive divider

The factor  $C_H/(C_H + C_A) \approx 0.667$  in Eq. 6 is not a refinement. Using a “simple”  $M_L = G M_c$  instead under-reads the pressure by  $1.5\times$ . Always use the parallel-circuit form, which is what the code does by default.

#### Common mistake — using a nominal sensitivity at a frequency where the calibration has structure

The 6 MHz dip in Figure 3a is about 20% deep and sits exactly where a harmonic-imaging pulse would land. Interpolate the certificate at the measured  $f_{\text{awf}}$ .

## 5.2 Manual calibration of $C_{\text{load}}$ for the no-preamp chain

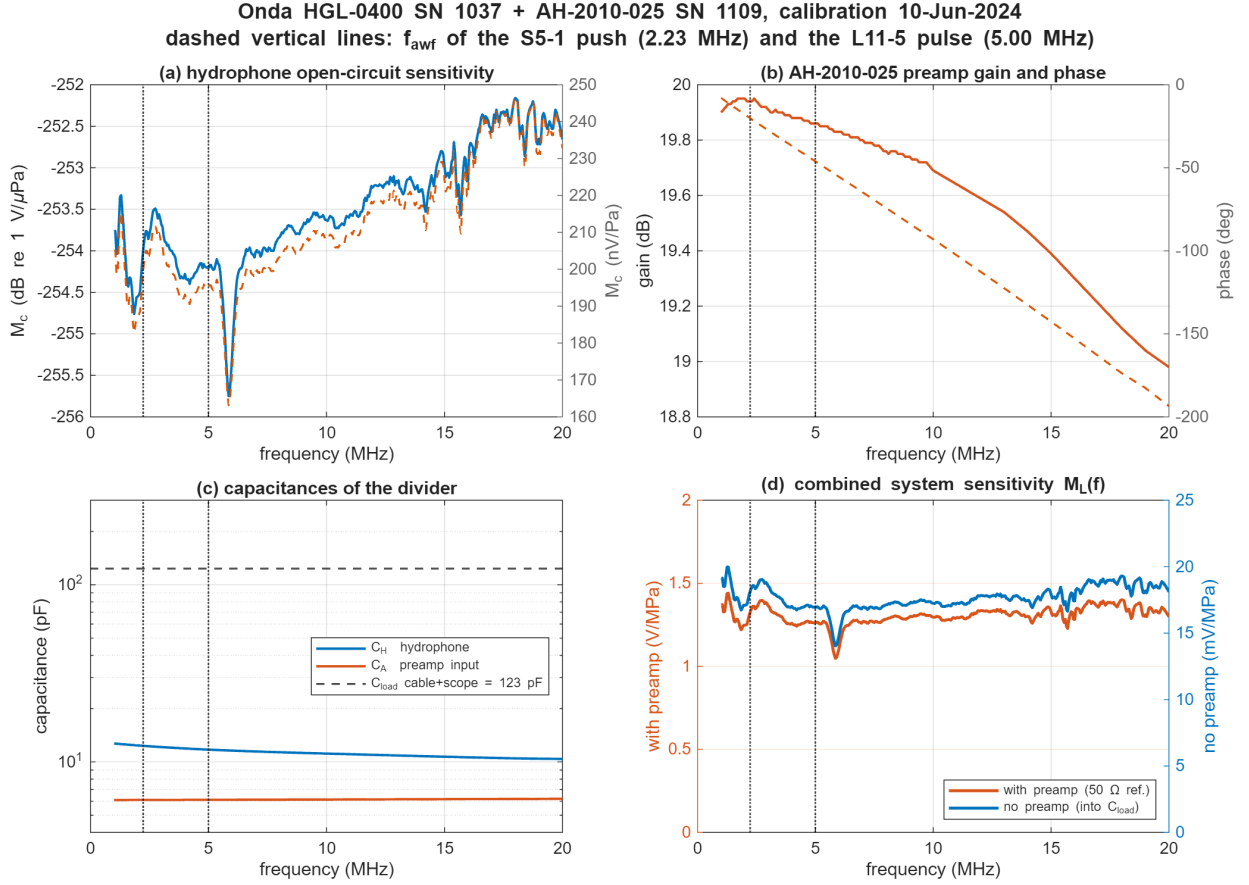
Because  $C_{\text{load}}$  dominates Eq. 7 and is a property of the specific cable and scope input rather than a specified value, it has to be measured. The two chains provide the means: at a fixed acoustic point, and at a drive where the pre-amplifier is unsaturated, their ratio depends only on known quantities and  $C_{\text{load}}$ ,

$$\frac{V_A}{V_B} = G \frac{C_H + C_{\text{load}}}{C_H + C_A} \implies C_{\text{load}} = \frac{V_A}{V_B} \cdot \frac{C_H + C_A}{G} - C_H. \quad (8)$$

### Procedure.

1. Park the hydrophone at a point with good signal — the field maximum is convenient — and do not move the stage, the probe or the tank for the rest of the step.
2. Choose several transmit voltages spanning the range where chain A is comfortably unsaturated (check the waveforms for flat tops).
3. At each voltage, capture with the pre-amplifier (into  $50 \Omega$ ), then remove the pre-amplifier and capture again (into  $1 \text{ M}\Omega$ ), changing nothing else.
4. Compute  $V_A/V_B$  at each voltage. **It must be constant.** Scatter beyond a few per cent means something moved, or chain A was already saturating.
5. Average the ratio over the unsaturated points and solve Eq. 8.

**Does this calibration transfer?**



**Figure 3:** The integrated calibration as a function of frequency. (a) Hydrophone open-circuit sensitivity  $M_c$ . (b) Pre-amplifier gain and phase; the phase matters only for a broadband deconvolution. (c) The three capacitances that set the dividers:  $C_H$  (hydrophone),  $C_A$  (preamp input, small — little loss) and  $C_{\text{load}}$  (cable+scope, ten times larger — most of the signal lost). (d) The combined system sensitivities of Eqs. 6 and 7 — note the different units on the two axes, a factor 73 apart. Dotted vertical lines mark the working frequencies of the two example measurements in §8.

- **Across probes and sequences: yes.**  $C_{\text{load}}$  is a property of the hydrophone cable and the scope input — the electrical side. It contains nothing about the transducer, the beam or the drive. Provided the same cable, adapters and scope input are used, the same  $C_{\text{load}}$  applies to any probe. (Determined twice, at different probes and different frequencies, agreeing to 2%; see §8. Those two determinations sample  $C_H(f)$  at 2.2 and 5.0 MHz, so their agreement is a genuine check on the divider terms  $C_H$ ,  $C_A$  and  $G$  — but not on  $M_c$ , which cancels out of the ratio.)
- **Across frequency: yes, within our band.** The divider  $C_H(f)/(C_H(f) + C_{\text{load}})$  depends on frequency only through  $C_H$ , which falls from  $\approx 12.7$  pF to  $\approx 10.5$  pF between 1 and 20 MHz — a couple of per cent on the divider. Cable capacitance is essentially frequency independent well below the cable’s self-resonance.
- **Re-determine it whenever the electrical path changes.** A different cable length, an added adapter or BNC tee, or a different scope input all change  $C_{\text{load}}$  directly. It is a five-minute measurement; repeat it rather than assume.

### 5.3 Acoustic working frequency

$f_{\text{awf}}$  enters MI as  $1/\sqrt{f}$  and the derating linearly, so it is worth getting right. Following IEC 62359, take the **arithmetic centre of the  $-6$  dB band of the fundamental** in the magnitude spectrum of the received waveform (Figure 4c).

$f_{\text{awf}}$  is generally *not* the programmed transmit frequency: it is the transmit spectrum multiplied by the probe’s transfer function, and the shift is largest for short, broadband pulses, which the probe passband pulls toward its own centre.

**Common mistake — measuring  $f_{\text{awf}}$  at high drive, or over the full band**

As the drive rises, nonlinear propagation builds harmonics, and a wideband centroid then drifts upward with transmit voltage — so the “same” sequence appears to change frequency. Restrict the band to the fundamental lobe, and take  $f_{\text{awf}}$  from a **low-drive** capture.

## 5.4 Derating

The regulated quantities describe what a target *in tissue* would experience, but the measurement is in water, which is essentially lossless. The standard remedy is a fixed homogeneous-tissue derating of  $0.3 \text{ dB cm}^{-1} \text{ MHz}^{-1}$  applied over the measured propagation distance  $z$  of Eq. 4:

$$p_{.3}(z) = p(z) \cdot 10^{-0.3 z f_{\text{awf}}/20}, \quad z \text{ in cm, } f_{\text{awf}} \text{ in MHz.} \quad (9)$$

Intensities derate with the *square* of that factor, since  $I \propto p^2$ . Derating is therefore a depth-dependent tilt on the measured profile, and **it is what makes the reportable maximum move** — see §6.3.

## 5.5 Pressure, pulse duration and the intensities

With  $M_L$  evaluated at  $f_{\text{awf}}$  and the load correction of Eq. 5 applied, the recorded trace becomes a pressure waveform,  $p(t) = V(t)/M_L$ , and

$$p_r = -\min_t p(t) \quad (\text{peak rarefactional pressure}), \quad (10)$$

$$I(t) = \frac{p(t)^2}{\rho c}, \quad \rho = 1000 \text{ kg m}^{-3}, \quad c = 1500 \text{ m s}^{-1}, \quad (11)$$

$$\text{PD} = 1.25 (t_{90} - t_{10}), \quad (12)$$

$$\text{PII} = \int_{t_{10}}^{t_{90}} I(t) dt, \quad (13)$$

where  $t_{10}$  and  $t_{90}$  are the times at which the cumulative intensity integral reaches 10 % and 90 % of its final value; the 1.25 factor is the convention that converts that 10–90 % span into an equivalent pulse duration. MI,  $I_{\text{sppa}.3}$  and  $I_{\text{spta}.3}$  then follow from Eqs. 1–3, with  $\text{PII}_{.3}$  the PII computed from the derated pressure. Figure 4 shows every one of these read off a single capture.

**Common mistake — taking  $p_r$  as half the peak-to-peak**

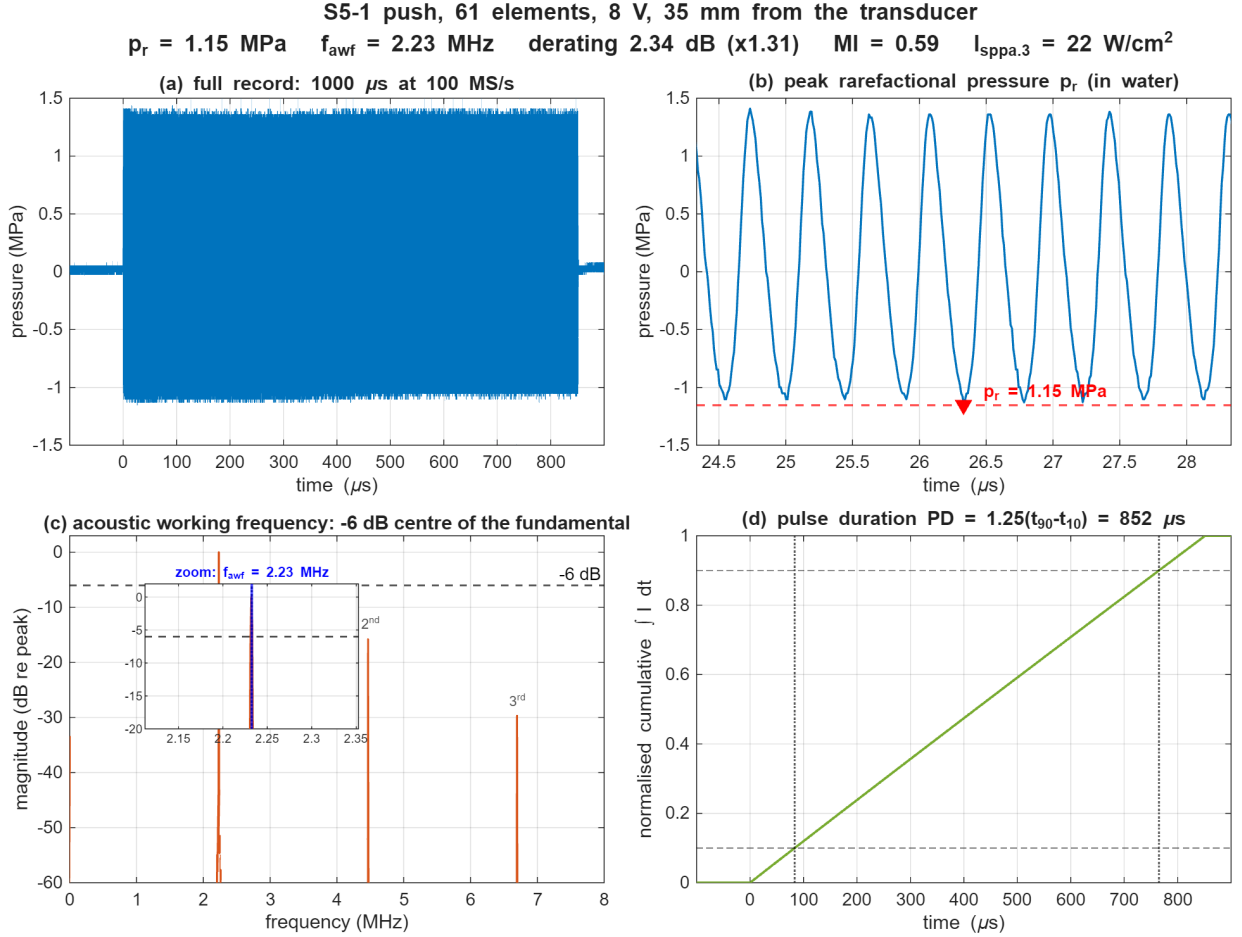
At the pressures involved, propagation is nonlinear and the waveform is asymmetric — compressional peaks are sharper and taller than the rarefactional troughs. MI uses the *rarefactional* half specifically. Take the minimum of the trace, never  $V_{\text{pp}}/2$ .

**Common mistake — using the line PRF for  $I_{\text{spta}}$**

The PRF in Eq. 3 is the **local** one: how often *that point in space* is insonified. For a scanned sequence that is the frame rate, not the line rate — a sequence firing 71 lines at  $270 \mu\text{s}$  PRI insonifies any given point once per 19.2 ms frame, i.e. at 52 Hz and not 3.7 kHz, a factor 71. For a burst-mode push, use the true duty cycle over the repeat period, including the off time.

## 5.6 Refinements that are usually not needed

Two corrections are available and both are small, in the direction of *increasing* the reported values:



**Figure 4:** Everything extracted from one saved waveform (S5-1 push, 61 elements, 8 V, 35 mm from the transducer). (a) The full 1000  $\mu$ s record; the 1900-cycle burst fills 850  $\mu$ s of it. (b) A few cycles around the negative peak; the waveform is visibly asymmetric — peaks at +1.40 MPa, troughs only at -1.15 MPa — which is nonlinear propagation. (c) Magnitude spectrum: the -6 dB centre of the fundamental gives  $f_{awf}$  (inset), with second and third harmonics at -16 and -30 dB. (d) Cumulative intensity integral, from which the pulse duration and PII follow.

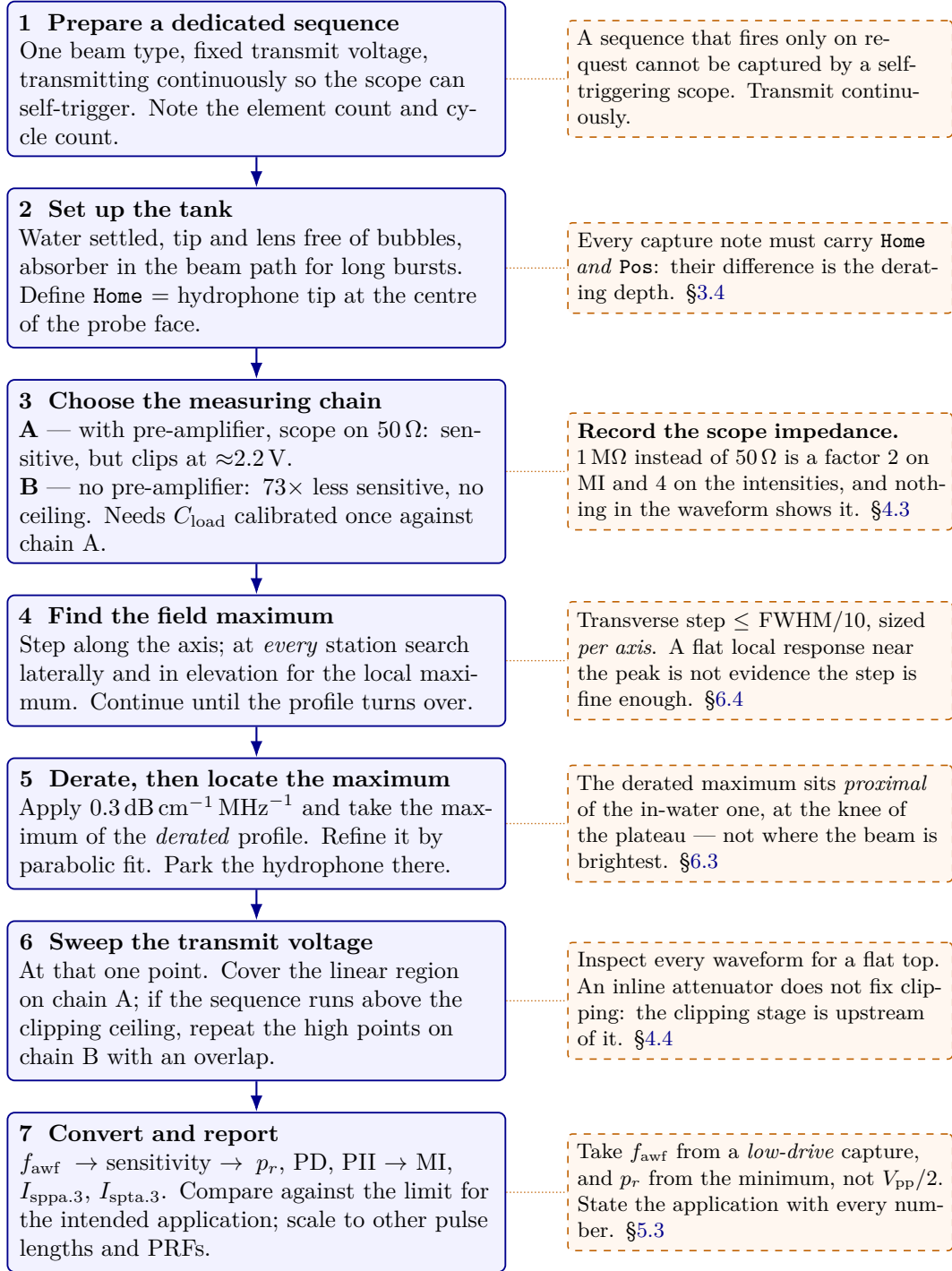
- **Spatial averaging.** The 400  $\mu$ m element averages the pressure over its face and so under-reads a tight focus. A jinc focal-plane model (`spatialAvgFactor.m`) gives a correction of a few per cent for our geometries.
- **Broadband deconvolution.** Instead of dividing by  $M_L(f_{awf})$ , deconvolve the full spectrum with  $M_L(f)$  including the amplifier phase (`voltageToPressureBroadband.m`). This changes MI by a few per cent, more at high drive where harmonics matter.

The single-frequency method is the default because it is what the standard prescribes and what reference values use. Since both corrections raise the reported values, leaving them out is *not* conservative — quote them as an uncertainty rather than ignoring them.

## 6 Measurement protocol

### 6.1 Step 1 — prepare a dedicated sequence

Do not measure the clinical sequence directly. Write a variant that transmits a **single, centred, non-steered beam**, repeated continuously at a convenient rate. The beam itself can be whatever the sequence uses — tightly focused, diverging, or anything in between — but it must be one beam, on the array centre, unsteered:



**Figure 5:** The measurement in seven steps (blue), each with the thing most easily got wrong in it (orange). Steps 4 and 5 are the ones that consume the time; step 3 is the one that silently invalidates everything downstream if the scope impedance is not recorded.

- **Centred and unsteered**, so the beam axis is nearly normal to the probe face; the axial sweep is then a straight line in stage coordinates and the transverse search stays small.
- **A single beam**, so the hydrophone is not sampling a different transmit on every trigger. Under a scanned sequence each line is seen once per frame, which mixes geometries and makes the local PRF ambiguous.
- **Repeated continuously**, so the scope can free-run on its own trigger.

Keep the transmit *waveform* — frequency, cycles, apodisation, aperture, focal depth — identical to the sequence being certified; those are what set the field. The PRF may differ, since  $I_{\text{spta}}$  is scaled analytically afterwards (§6.6).

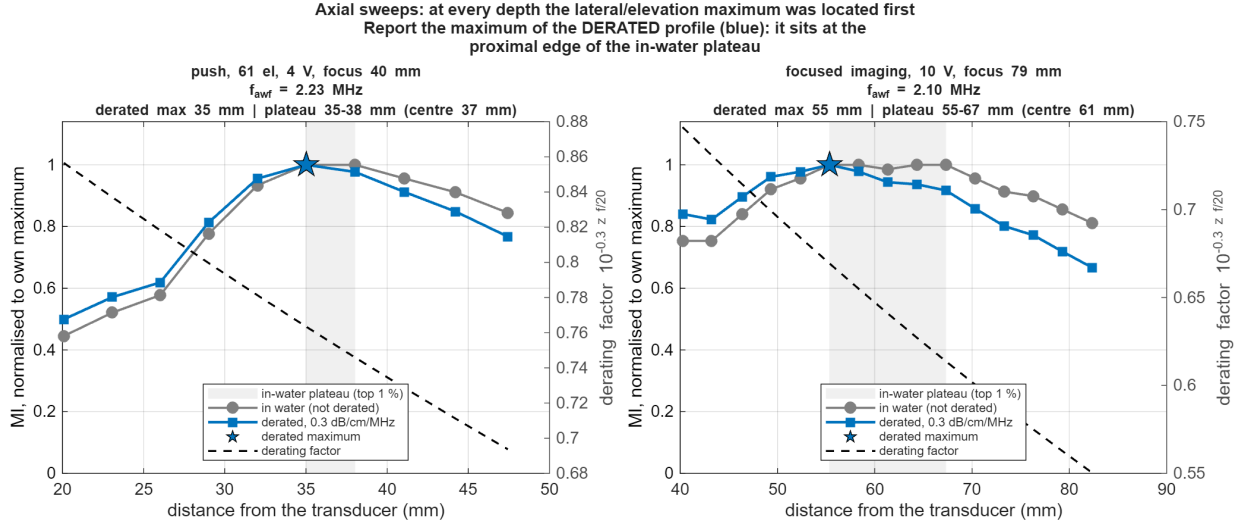
## 6.2 Step 2 — set up and check

1. Mount probe and hydrophone facing each other, wipe both surfaces, and check for bubbles.
2. Define **Home** at the transducer face centre (§3.4).
3. Verify the 2 : 1 impedance ratio (§4.3).
4. Run the reflection check of §3.3 if the pulse is long, and add absorber if the burst envelope is not flat.
5. Determine  $C_{\text{load}}$  (§5.2) if chain B will be used.
6. Capture at low drive and confirm  $f_{\text{awf}}$  (§5.3).

## 6.3 Step 3 — find the maximum of the derated field

The regulated quantity is the maximum of the **derated** field, and **its location is not known until the data have been analysed**. Three effects displace it from the nominal focus: diffraction (focal shift, proximal), nonlinear propagation (further proximal), and derating, which grows with depth and therefore **always pulls the maximum proximal**. Probe–hydrophone misalignment displaces it laterally and in elevation. Hence:

1. **Start conservatively shallow**. Use a field simulation to bracket where the in-water focus should be, then **begin proximal of it**, because the derated maximum is on that side.
2. **Step along the axis**. A uniform 3 mm step is a reasonable default. A two-stage alternative is a coarse 5 mm map followed by a 2 mm refinement within  $\pm 10$  mm of the observed maximum. As a rule of thumb the  $-6$  dB depth of focus is  $\approx 7.1 F_{\#}^2 \lambda$ ; take a step of roughly a tenth of that.
3. **At every axial station, search transversely**. Move in lateral and elevation to find the local maximum of  $|V_{\text{neg}}|$  before saving. This is what corrects the probe–hydrophone misalignment, and **the step size here is the single most under-estimated parameter in the whole measurement** — see §6.4. Work in two passes: a coarse pass at  $\approx \text{FWHM}/3$  over  $\pm 1$  FWHM to locate the lobe, then a **fine pass at  $\approx \text{FWHM}/10$** , which for our probes means 0.1 mm to 0.2 mm.
4. **Save only the local peak waveform, with its coordinates**, in the note format of Table 3. One sweep then serves MI and both intensities.
5. **Continue until the profile clearly turns over** and stays down for several stations, so the maximum is genuinely bracketed.
6. **Analyse before concluding**. Apply the derating and take the maximum of the derated profile — not of the raw one — then refine it by parabolic interpolation and check that the axial step was fine enough (§6.4).



**Figure 6:** Why the derated maximum has to be found rather than assumed. Two axial sweeps, each point already the local lateral/elevation maximum. Grey: in water. Blue: after derating. Shaded: the depths within 1% of the in-water maximum. Dashed (right axis): the derating factor, which falls monotonically with depth and tilts the profile proximal. *Left:* a shallow 40 mm-focus push, where the shift is modest. *Right:* a 79 mm-focus imaging beam, whose in-water field is a 12 mm-wide plateau and whose *derated* maximum sits at the shallow edge of it, 24 mm proximal of the nominal focus. The plateau is **real, not a sampling artefact**: at  $F_{\#} = 3.9$  and 2.1 MHz the  $-6$  dB depth of focus is  $7.1 F_{\#}^2 \lambda \approx 77$  mm, so the field genuinely varies by less than a per cent over the 12 mm shaded span. What the 8-bit digitizer contributes is only that neighbouring stations then return *bit-identical* codes (one code is 0.7% here), which makes the in-water maximum unlocatable — while the derated maximum stays sharp, for the reason given in §6.4.

#### Common mistake — stopping the sweep at the nominal focus

Because derating tilts the profile proximal, the derated maximum can sit well shallower than the in-water focus — particularly for a deep focus, where the in-water field is a long plateau and the derating gradient across it is large. A sweep that stops at, or starts at, the nominal focus can miss it entirely and under-report MI substantially.

#### Common mistake — skipping the transverse search

An uncorrected elevation offset produces a *constant multiplicative* under-read: the shape of every curve is right, the slope of MI against transmit voltage is right, and only the absolute value is wrong. **A constant factor across a voltage sweep is a positioning error, not a calibration error** — and if only one location is ever measured, the two are indistinguishable.

### 6.4 Choosing the step sizes

Both the axial sweep and the transverse search at each station need a step small enough that the recorded maximum really is the maximum.

**Transverse (lateral and elevation).** Size the step as a **fraction of the beam width**, never as an absolute distance. Modelling the focal-plane amplitude as a sinc of half-width FWHM =  $1.21 \lambda F_{\#}$ , a square grid of step  $\Delta$  can miss the peak by  $\Delta/2$  in both axes at once, costing  $1 - \text{sinc}^2(\pi\Delta/2)$  in amplitude. A step of FWHM/3 costs up to 13% on MI and 24% on the intensities; FWHM/10 costs about 1%.

Use  $\Delta \leq \text{FWHM}/10$ , recomputed for every geometry, and sized *per axis* — elevation is focused by a fixed lens, so its width differs from the lateral one and the narrower of the two governs its own step.

Going finer than FWHM/10 buys little: at 1 % the error is already well below the hydrophone calibration uncertainty, and the 8-bit digitizer's own quantisation floor is a few tenths of a per cent. For our geometries this works out at 0.1 mm to 0.2 mm for the S5-1 and about 0.05 mm for the L11-5 at 5 MHz — tight enough that stage resolution, not patience, becomes the limit, which is a further argument for the automated raster of §2.1.

**Axial.** A few millimetres is enough. The derated profile falls monotonically across the in-water plateau (a flat function times a monotonically decreasing one), so its maximum sits at the sharp *knee* where the rising flank meets the plateau, not somewhere unlocatable in the middle. 3 mm resolved that knee in both of our sweeps. Confirm it rather than assume it: fit a parabola through the sampled maximum and its two neighbours — near-symmetric neighbours mean the step was fine enough, and the fitted vertex gives sub-step localisation for free, while strongly asymmetric ones are the signal to refine.

**Common mistake — concluding from a flat local response that the step is fine enough**

Near the peak the response is genuinely flat — a 0.1 mm move changes the reading by only 0.5 %, which is easy to read as “precise enough”. The trap is that the same move costs 7.6 % at 0.6 mm off axis and 14.5 % at 1.0 mm. **Large sensitivity to a small move is the diagnostic that you are on the flank, not at the peak** — keep refining until the response goes flat, and only then record.

## 6.5 Step 4 — sweep the transmit voltage at that point

Park the hydrophone at the derated maximum and sweep the transmit voltage; this is the measurement that becomes a permissible-voltage limit. Cover the linear region on chain A, and if the sequence will be used above the clipping ceiling, repeat the high-voltage points on chain B, overlapping the two chains over several voltages.

## 6.6 Step 5 — what to measure, and what to scale

Re-measuring the whole field for every sequence variant is unnecessary:

- **Changes the field shape → measure.** Beam type (focused / diverging / plane), aperture or element count, focal depth, transmit frequency. Each needs its own axial sweep, because each moves the maximum.
- **Scales predictably → measure once at the peak and scale.** Transmit voltage ( $MI \propto V$ , intensities  $\propto V^2$ , up to nonlinear saturation — verify with a few points across the range); pulse length (scales  $I_{\text{spta}}$  only); PRF and duty cycle (a global multiplier on  $I_{\text{spta}}$ ).

**Common mistake — treating the long-time-average  $I_{\text{spta}}$  as the whole thermal story**

A burst duty cycle — a second of pushing followed by tens of seconds of quiet — makes the regulatory  $I_{\text{spta}}$  small, and that is legitimate. But the transient temperature rise *within* each burst is a separate question that  $I_{\text{spta}}$  does not answer. Check the thermal indices, and probe-face heating, separately.

# 7 Checklist

Run through this before starting, and again before reporting. The four groups below correspond to the four places a measurement usually goes wrong: the electrical chain, the digitizer, the positioning, and the reporting.

---

|                                                                             |                                                                                |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Electrical chain</b>                                                     |                                                                                |
| Scope impedance recorded, 2:1 ratio verified                                | A factor 2 on MI, 4 on the intensities, invisible in the waveform (§4.3).      |
| Waveforms inspected for flat tops                                           | Clipping is not obvious from the peak value alone (§4.4).                      |
| No attenuator used to “fix” clipping                                        | The clipping stage is upstream of it; remove the pre-amplifier instead (§4.4). |
| $C_{\text{load}}$ re-determined after any cabling change                    | It dominates the no-preamp reading and is not a datasheet value (§5.2).        |
| <b>Digitizer</b>                                                            |                                                                                |
| Real-time sampling, not RIS                                                 | Equivalent-time sampling is invalid for a single-shot push (§4.6).             |
| Record length > pulse length                                                | A truncated record silently under-reports PII and both intensities.            |
| Vertical range re-checked after each voltage change                         | 8-bit digitizer; a loose range throws away resolution.                         |
| Full waveform saved, not just $V_{\text{pp}}$                               | Everything except $p_r$ needs the waveform (§4.5).                             |
| <b>Positioning</b>                                                          |                                                                                |
| Home and Pos in every capture note                                          | Sets the derating depth; 5 mm of origin error is already 0.3 dB (§3.4).        |
| Transverse search at <i>every</i> axial station, step $\leq \text{FWHM}/10$ | Otherwise a constant multiplicative under-read (§6.3, §6.4).                   |
| Axial sweep continued until the profile turns over                          | The derated maximum is proximal of the in-water one (§6.3).                    |
| Bubbles checked on tip and lens; absorber in place for long bursts          | Both read low, and neither is visible in the number (§3.2, §3.3).              |
| <b>Analysis and reporting</b>                                               |                                                                                |
| $f_{\text{awf}}$ from a low-drive capture, fundamental band only            | Harmonics inflate a wideband estimate as drive rises (§5.3).                   |
| $p_r$ from the minimum, not $V_{\text{pp}}/2$                               | The waveform is nonlinearly asymmetric (§5).                                   |
| Local (per-point) PRF used for $I_{\text{spta}}$                            | For a scanned sequence that is the frame rate, not the line rate.              |
| Application stated with every reported number                               | $I_{\text{spta},3}$ ranges over $40\times$ between applications (§1.1).        |
| Thermal indices checked separately for burst sequences                      | $I_{\text{spta}}$ does not capture the within-burst transient (§6.6).          |

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## 8 Worked results

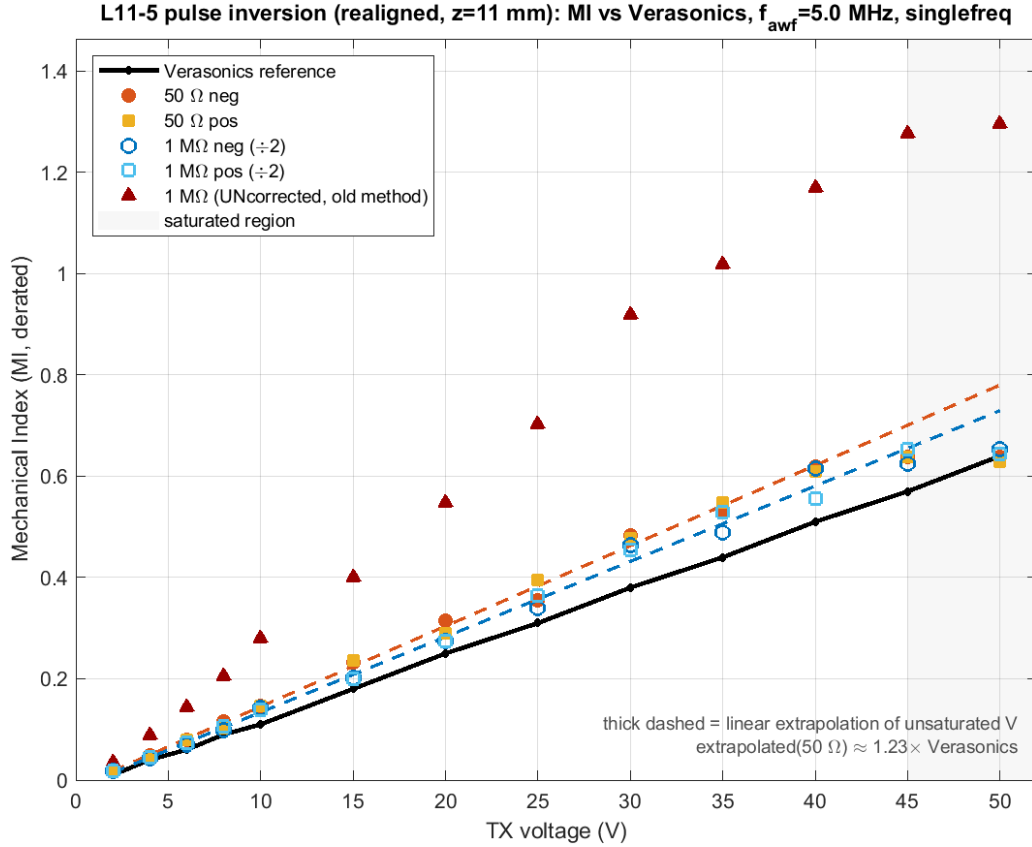
This section reports our own measurements, as examples of what the protocol produces.

### 8.1 Validation against Verasonics on the L11-5

Verasonics reports MI at a single location for a small number of its example scripts — far from all of them, but enough to validate the method against an independent reference. We used L11-5v High Image Quality (WideBeamHISC), a harmonic pulse-inversion sequence transmitting one cycle at 4.46 MHz, received at  $f_{\text{awf}} = 5.0$  MHz, and measured at the reference depth of 11 mm over a 2 V to 50 V transmit sweep, in all four combinations of pulse polarity and scope impedance.

Figure 7 validates four things at once:

- **The impedance correction is right.** The  $50\ \Omega$  and  $1\ \text{M}\Omega \div 2$  series coincide, while the uncorrected  $1\ \text{M}\Omega$  points sit a clean factor 2 high.
- **Normal and inverted pulses agree**, as they must for pulse inversion.



**Figure 7:** L11-5 pulse inversion at 11 mm: measured MI versus transmit voltage against the Verasonics reference (black). Circles and squares: the corrected analysis at  $50\ \Omega$  and at  $1\ \text{M}\Omega$  with the  $\div 2$  load correction — the four series coincide. Red triangles: the same  $1\ \text{M}\Omega$  data *without* the load correction, running away to  $2\times$ . Shaded: where the pre-amplifier saturates; dashed lines are linear extrapolations from the unsaturated points.

- **The shape is right:** MI is linear in transmit voltage with the same slope as the reference, over the whole unsaturated range.
- **The absolute value agrees to  $1.23\times$ ,** roughly constant with voltage, after extrapolating the unsaturated ( $\leq 30\ \text{V}$ ) region.

**On the residual  $1.23\times$ .** We could not remove it, having excluded: using the programmed  $4.46\ \text{MHz}$  rather than the measured  $5.0\ \text{MHz}$  (which makes it worse,  $1.32\times$ ); spatial averaging by the  $400\ \mu\text{m}$  tip (a few per cent, in the wrong direction); broadband deconvolution (also the wrong direction); tank reflections (the echo at  $2 \times 11\ \text{mm}/c \approx 14.7\ \mu\text{s}$  falls outside the  $5\ \mu\text{s}$  window); bubbles (they attenuate, and erratically, whereas this is a steady over-read); and the derating depth. What remains is the absolute accuracy of the hydrophone calibration ( $\pm 1\text{--}2\ \text{dB}$  is typical) and whatever model Verasonics uses;  $1.23\times$  is  $1.8\ \text{dB}$ , within the combined uncertainty of the two. **Note the direction:** we measure *higher* than Verasonics, which is conservative for safety but means this agreement cannot be used to argue our numbers are pessimistic.

**Why the transverse search is in the protocol.** An earlier version of this dataset, taken without re-aligning in elevation, gave MI a constant  $0.36\times$  Verasonics — a pure multiplicative offset with the shape preserved, exactly the signature described in §6.3. Re-aligning produced Figure 7.

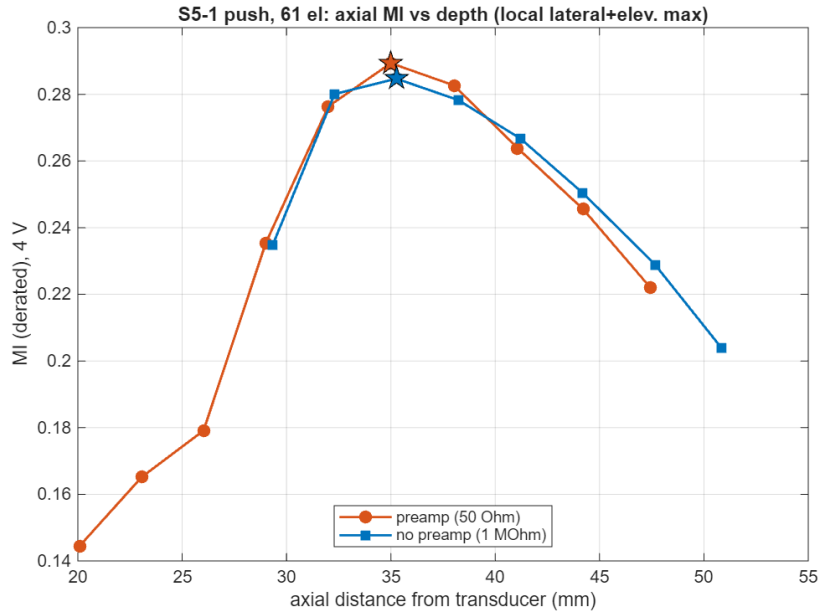
**Table 5:** Maximum permissible transmit voltage for the S5-1 push at the derated field maximum ( $z \approx 36$  mm, 1900 cycles, burst duty 0.77 Hz effective), from the chain-B measurements, requiring *both*  $MI \leq 1.9$  and  $I_{\text{sppa.3}} \leq 190 \text{ W cm}^{-2}$ .

| Aperture    | $V$ at $MI = 1.9$ | $V$ at $I_{\text{sppa.3}} = 190$ | $V$ at $I_{\text{spta.3}} = 720$ | <b>max</b> $V$ | Binding limit       |
|-------------|-------------------|----------------------------------|----------------------------------|----------------|---------------------|
| 41 elements | 39.3 V            | 32.0 V                           | $> 50 \text{ V}$                 | <b>32 V</b>    | $I_{\text{sppa.3}}$ |
| 61 elements | 27.9 V            | 23.1 V                           | $> 50 \text{ V}$                 | <b>23 V</b>    | $I_{\text{sppa.3}}$ |
| 79 elements | 23.1 V            | 20.1 V                           | $> 50 \text{ V}$                 | <b>20 V</b>    | $I_{\text{sppa.3}}$ |

## 8.2 S5-1 ARF push

The target measurement: the focused ARF push used for shear-wave elastography on the S5-1 (1.95 MHz programmed,  $f_{\text{awf}} = 2.23$  MHz, focus 40 mm, 1900 cycles, apertures of 41 / 61 / 79 elements, burst duty of 24 pushes at 20 Hz followed by  $\geq 30$  s off, i.e. 0.77 Hz effective). No reference value exists for it anywhere.

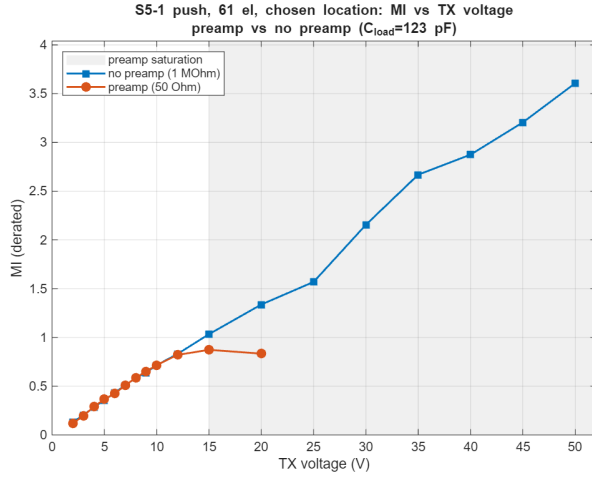
**Locating the maximum.** The axial sweep (Figure 6, left) was run at 4 V in 3 mm steps from 20 mm to 47 mm, with a transverse search at every station. The derated maximum falls at  $z \approx 35$  mm, proximal of the 40 mm nominal focus, and is bracketed. Figure 8 repeats the sweep on both chains: the two — a factor 73 apart in sensitivity — agree on both the profile and the location of the maximum, which is a useful end-to-end check.



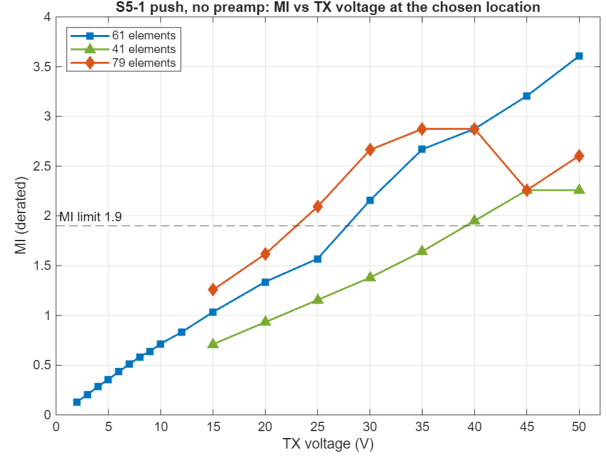
**Figure 8:** The same axial sweep measured on both chains.

$C_{\text{load}}$ . The procedure of §5.2 gave  $C_{\text{load}} = 123.3 \text{ pF}$ , constant over 3 V to 12 V transmit. An independent determination in the L11-5 session — different probe, different frequency — gave 121 pF, a 2% agreement, which is the evidence behind the transferability claim in §5.2.

**Voltage and aperture.** At the peak point the transmit voltage was swept from 2 V to 20 V on chain A and 2 V to 50 V on chain B (Figure 9a), and the sweep repeated at 41, 61 and 79 elements (Figures 9b and 10). Chain A is usable to  $\approx 10$  V; chain B runs to the full 50 V and agrees with chain A where they overlap.



(a) MI versus transmit voltage on both chains. The shaded region is where chain A saturates.

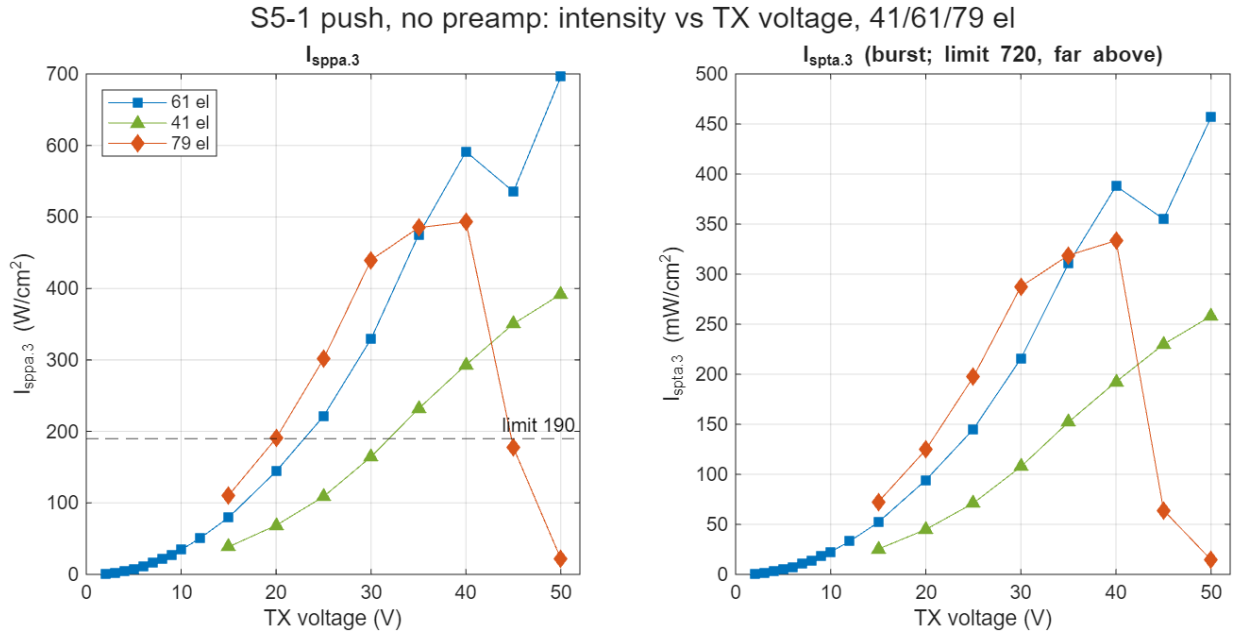


(b) MI versus transmit voltage for the three apertures, chain B. MI crosses 1.9 at 39 V (41 el), 28 V (61 el) and 23 V (79 el).

**Figure 9:** S5-1 push voltage sweeps at the derated field maximum. **The non-monotonic points above  $\approx 40$  V** are single captures in a strongly shocked, supply-sagging regime and are not reliable; they lie far above every limit crossing.

**Outcome.** Table 5 gives the crossings. Four observations that generalise:

- **$I_{\text{sppa.3}}$  binds before MI** for these long focused pushes: at the permissible voltage MI is only  $\approx 1.5$ – $1.6$ . Checking MI alone — the index everyone quotes — overshoots by 15 % to 25 % in voltage. (Under a strict reading of the **or** in Table 2 the MI crossing would be the operative one; we require both.)
- **The burst duty is what buys the  $I_{\text{spta}}$  headroom.** At the permissible voltages  $I_{\text{spta.3}}$  sits near  $100 \text{ mW cm}^{-2}$  — under both the 720 and the cardiac  $430 \text{ mW cm}^{-2}$  limit. Change the duty cycle and this is the first number to recompute.
- **Extrapolation from an unsaturated region works.** A separate analysis fitting only the low-voltage chain-A sweep and extrapolating ( $\text{MI} \propto V$ ,  $I_{\text{sppa}} \propto V^2$ ) predicted 20.0 V for 79 elements and 24.3 V for 61 — within 6 % of the directly measured values. It worked here because the limits are crossed at 20 V to 32 V, below where saturation bites.
- **But the two chains are not independent, and this is not an absolute check.**  $C_{\text{load}}$  was solved from the chain-A/chain-B voltage ratio (§5.2), so chain B’s pressure scale is derived from chain A’s calibration. An error in the preamp-chain sensitivity — above all in the hydrophone sensitivity  $M_c$ , the one term carrying 1 dB to 2 dB of uncertainty — propagates into  $C_{\text{load}}$  and then into every no-preamp number, by the same factor, and cannot show up in this comparison. What the agreement does establish is that the extrapolation is sound and that neither chain is distorted by saturation at the crossing voltages. **Testing the absolute scale needs a second, independently calibrated hydrophone** — see §8.1, where the residual  $1.23\times$  against Verasonics is exactly the kind of error this comparison is blind to.
- **The table is deliberately pessimistic.** It is built from the peak-optimised single captures at each voltage, which read higher than repeated captures at the same nominal position (102 mV versus 66 mV at 40 V, 79 elements). That is the correct conservative choice for a limit, but it should not be quoted as the typical output.



**Figure 10:** The two intensities for the same three apertures. *Left:*  $I_{sppa.3}$  crosses  $190 \text{ W cm}^{-2}$  at 20 V to 32 V. *Right:*  $I_{spta.3}$  at those voltages is only  $100 \text{ mW cm}^{-2}$  to  $190 \text{ mW cm}^{-2}$ , entirely because of the burst duty cycle. Above  $\approx 40 \text{ V}$  the 10–90% window that defines the pulse duration becomes unstable on a shocked waveform — at 45 V it returns  $470 \mu\text{s}$  for an  $850 \mu\text{s}$  burst — which is what produces the apparent collapse of the 79-element curve.

## 9 Code and data

Everything this document refers to — the code, the calibration, and the raw `.hws` captures behind every figure and every number in §8 — is in the repository this report lives in. There is no external dependency and no absolute path: clone it, run `misetup`, and every script finds its own data. The one exception is Appendix `efapp:probes`, whose two figures come from a separate phantom shear-wave analysis in the processing repository; they are included here as images only.

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|                                       |                                                                     |
|---------------------------------------|---------------------------------------------------------------------|
| <code>docs/report/</code>             | this document and its figures                                       |
| <code>code/readHWS.m</code>           | read an <code>.hws</code> capture (waveform + config + note)        |
| <code>code/SafetyIndices.m</code>     | waveform $\rightarrow$ MI, $I_{sppa.3}$ , $I_{spta.3}$              |
| <code>code/SystemSensitivity.m</code> | calibration $\rightarrow \text{V MPa}^{-1}$ , either chain          |
| <code>code/analysis/</code>           | the full analyses, including <code>MakeReportFigures</code>         |
| <code>calibration/</code>             | the two Onda certificates (source of truth) and the digested tables |
| <code>data/</code>                    | the captures used here, with the acquisition note inside each file  |
| <code>demo/RunDemo.m</code>           | an end-to-end worked example that checks itself                     |

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To reproduce the figures of this report:

```
misetup
MakeReportFigures
```

`RunDemo` is the fastest way to see the whole chain work on real data; it ends by comparing every quantity it computed against stored reference values, so a broken installation announces itself rather than producing plausible wrong numbers.

## A Operating the OWIS motion stage

The hydrophone is carried by an OWIS three-axis motion stage (Figure 11), driven from OWISoft on the Vantage 256 PC. Every coordinate in this report — **Home**, **Pos**, the axial step, the transverse search — is a stage coordinate, so this appendix records how to drive it. It is a condensed version of the standalone note *OWIS motion stage — how to setup and use* (19 March 2024); the OWISoft and PS-10 manuals are the reference for anything beyond this, and are on the group NAS alongside the installer at `Ultrasound-BMd\Software\OWIsoft`.



**Figure 11:** The OWIS three-axis motion stage, with the hydrophone holder mounted vertically.

### A.1 Starting up

1. On the Vantage 256 PC, start **OWISoft** (Desktop → **OWIsoft** folder).
2. Open all three control units: **File** → **Control unit** → **Open**, and take **X**, **Y** and **Z** from the default folder.
3. **Tick *active* for all three** (Figure 12, left). An axis that is not activated silently ignores every move command.

### A.2 Free positioning — moving by hand

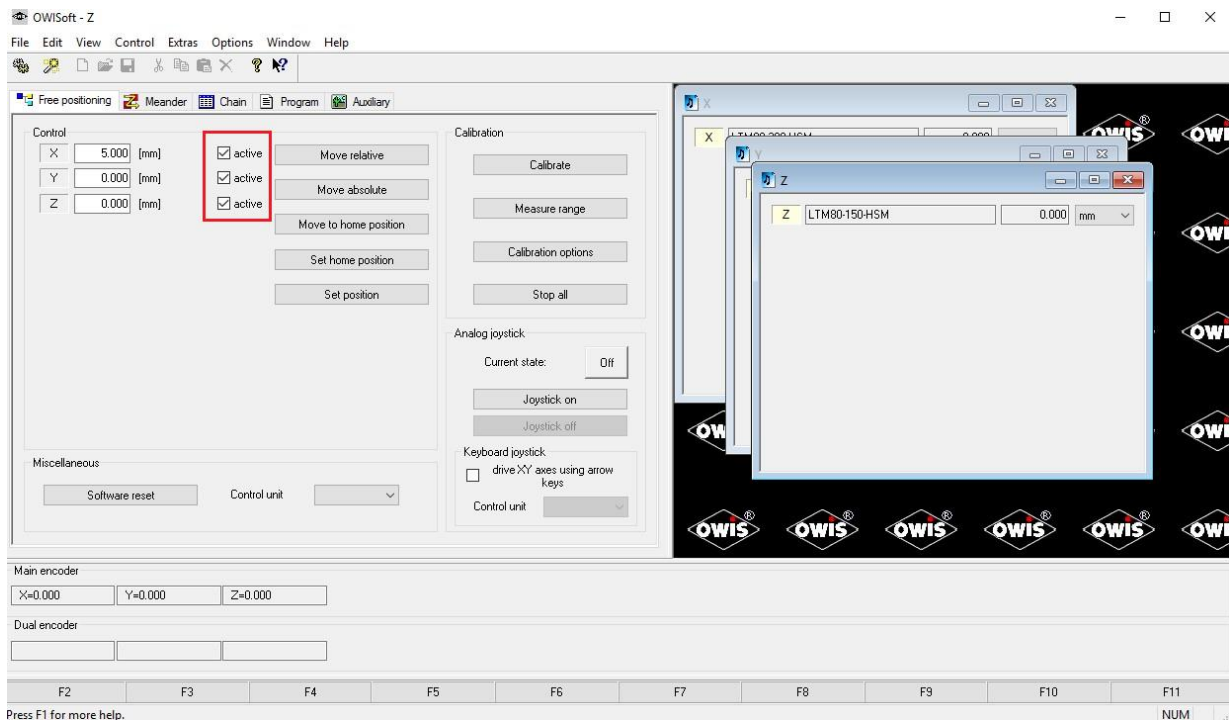
The **Free positioning** tab (Figure 12) is what the whole measurement of §6 is driven from. Enter a displacement per axis and press **Move relative**, or an absolute coordinate and **Move absolute**; several axes may move at once. In the example shown the stage would advance 5 mm in  $x$ .

**Use *Set home position once the hydrophone tip is at the centre of the transducer face***, and the stage will return there with *Move to home position*. That point is the **Home** of §3.4, and the derating depth of every capture is measured from it, so setting it deliberately at the start of a session is what makes the rest of the geometry meaningful.

### A.3 Meander — scanning a grid automatically

The **Meander** tab (Figure 13) cycles through a rectangular grid defined by a step count and a step length per axis, moving in  $x$  first, then  $y$ , then  $z$ . Between points it waits for a fixed delay, a mouse click, or the completion of a custom function.

This is the mechanism the automated 3-D raster of §2.1 would be built on. What is missing is not the motion — it is the synchronisation: the meander must trigger the digitizer and wait for a capture to be written before stepping on. The *wait for function* option is the hook for that.



**Figure 12:** Free positioning. Left: the three axes, each with its *active* checkbox — all three must be ticked. *Set home position* defines the Home used throughout this report.

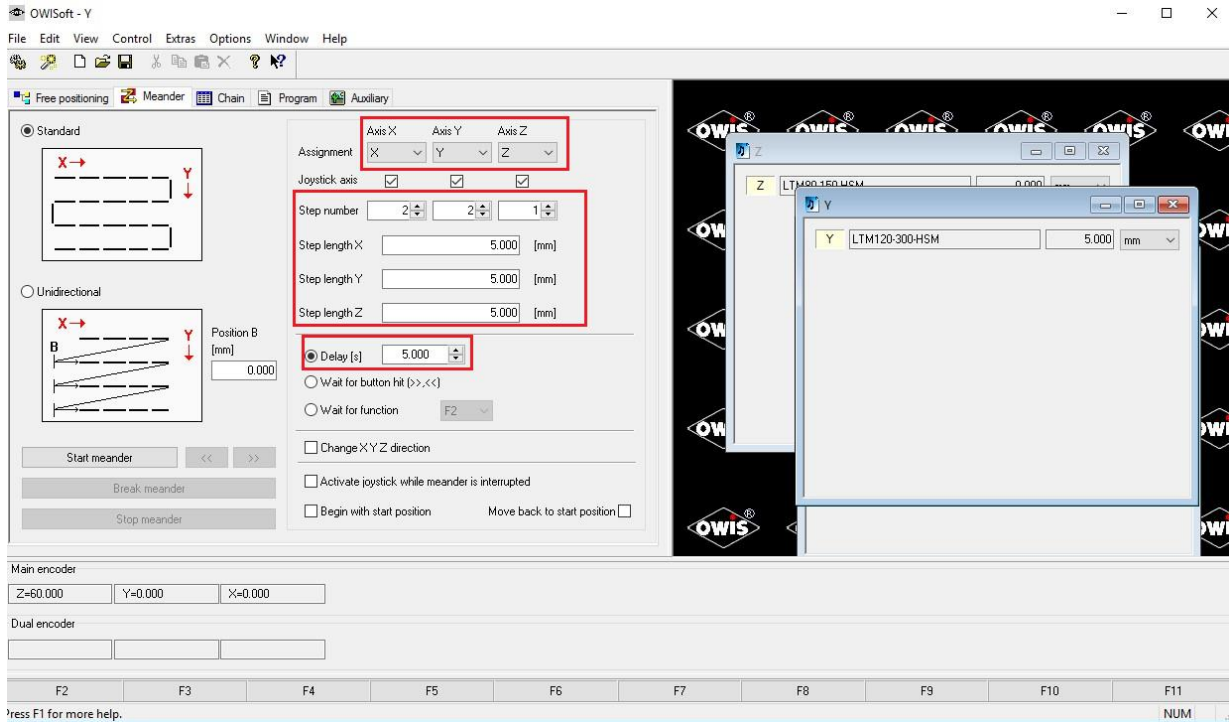
## A.4 First-time setup

*Reference only — this was done on the Vantage system on 19 March 2024 and does not need repeating unless the PC or the controllers are replaced.*

1. **Install OWISoft.** The installer and the OWIS manuals are on the group NAS at Ultrasound-BMd\Software they are licensed software and are deliberately not in this repository. Setup asks for a user name, a company and an installation password — **the password is not reproduced here**, it is with the installer on the NAS.
2. **Configure three control units**, one per axis: **File** → **Control unit** → **New** (Figure 14a), then
  - select the COM port — if unsure which is which, unplug the units one at a time and see which port disappears;
  - set **Control unit type** to PS10;
  - tick *Initialize axes automatically* and *Connect automatically at startup*;
  - press **Connect**.
3. An axis-configuration wizard opens (Figure 14b):
  - under **Step 1**, *Show dialog box*, and name the axis (axis 1 → X);
  - *Identify* finds the positioning-unit type number;
  - *Auto-configuration*, then close;
  - under **Step 5**, *Initialize*, then close.

Repeat for Y and Z.

4. **Save:** File → Control unit → Save.



**Figure 13:** Meander. The grid comes from the step count and step length per axis; the dwell between points is a fixed delay, a button press, or a custom function — the last being the hook a synchronised raster would use.

## B Are our two S5-1 probes equivalent?

*This appendix does not concern the hydrophone measurement. It is included here because the question — whether a safety measurement made on one of our two S5-1 probes transfers to the other — arises in the same conversations.*

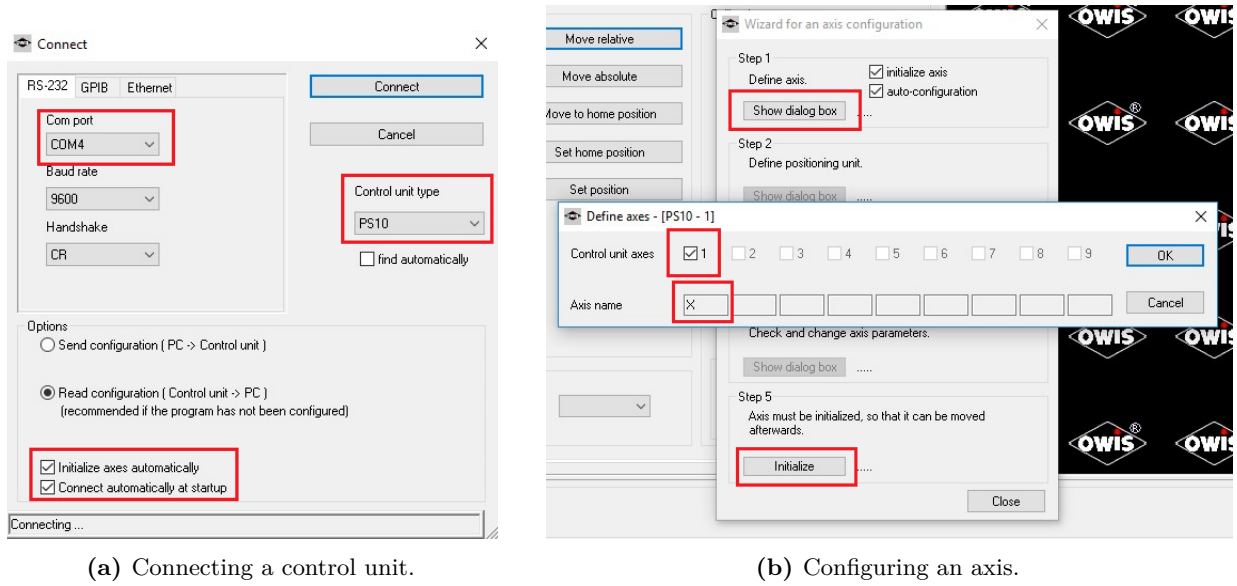
We have two S5-1 probes, the TU/e one and a second borrowed from MUMC. The suspicion worth eliminating was that the TU/e probe had degraded, which would both weaken our in-vivo push and invalidate any transfer of measurements between them.

**What was done.** On 2026-08-18 both probes swept the identical 3 apertures  $\times$  2 pulse lengths  $\times$  5 transmit voltages grid on the same CIRS phantom, within 45 min of each other, 10 pushes per cell, plus one focused B-mode of a resolution phantom each. Every cell was scored with the same shear-wave quality metric (mirror symmetry of the space-time plot). Crucially, the comparison was anchored against a **repeatability control**: the TU/e probe’s own results from the previous day, re-coupled.

**Result.** Pooling all 30 cells:

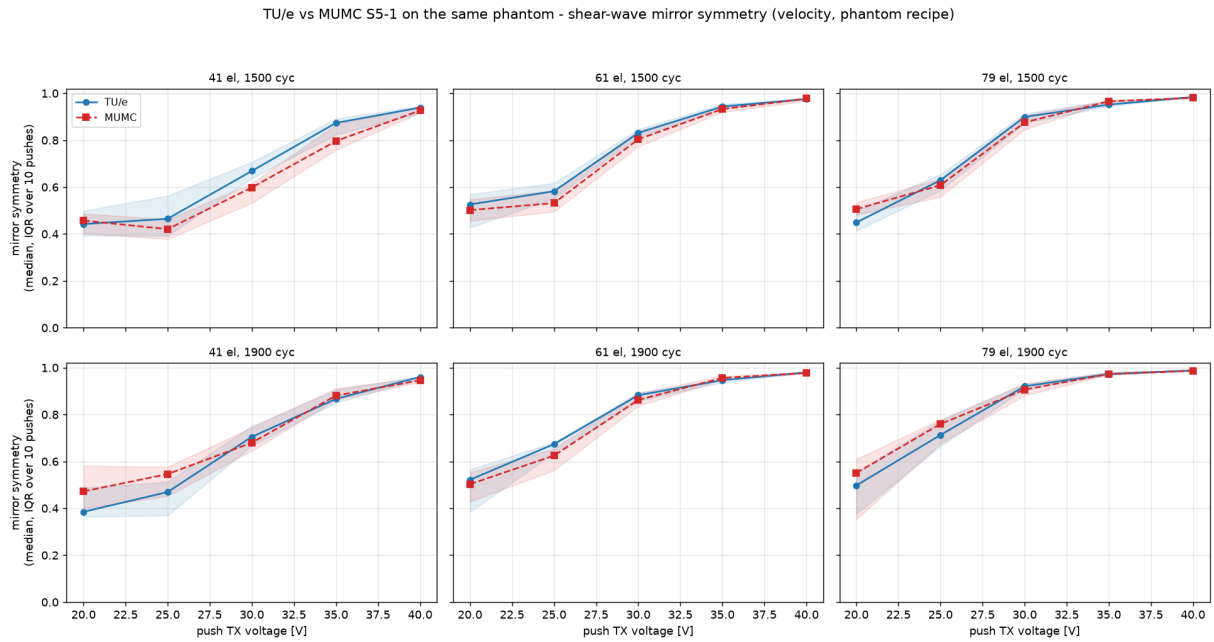
| Comparison                                               | mean   | median | RMS   | worst cell |
|----------------------------------------------------------|--------|--------|-------|------------|
| MUMC – TU/e (same day, same phantom)                     | −0.004 | −0.011 | 0.039 | 0.087      |
| TU/e 08-18 – TU/e 08-17 ( <i>same</i> probe, re-coupled) | +0.002 | +0.005 | 0.034 | 0.086      |

**The probe-to-probe difference is the same size as the day-to-day repeatability of a single probe** — i.e. it is measurement noise, and the two probes are indistinguishable by this test. The curves overlap within the inter-quartile range of the 10 pushes essentially everywhere (Figure 15), and the resolution-phantom B-modes agree as well (Figure 16); the apparent depth difference there is a lateral placement difference of the wire column, not sensitivity. As a bonus,



**Figure 14:** First-time setup of one axis. Repeat for all three.

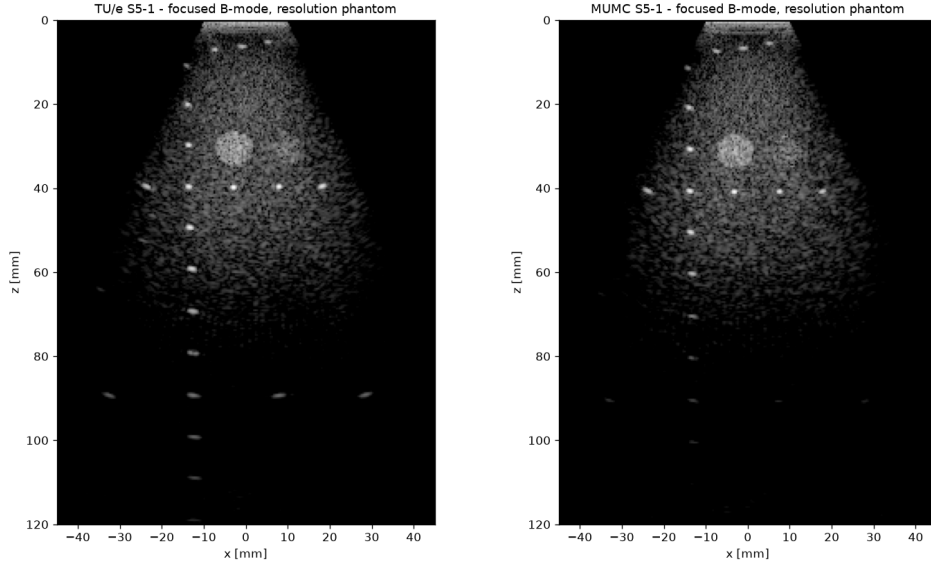
the dataset independently replicates the aperture/voltage ranking from the same campaign, to within 0.05 on both probes.



**Figure 15:** Shear-wave quality versus transmit voltage for the TU/e and MUMC S5-1, one panel per aperture and pulse length, same phantom, same day. Shaded bands are the inter-quartile range over 10 pushes.

### What this does and does not license.

- It **does** rule out the transducer as the explanation for the weak in-vivo push, which is the question it was designed to answer.
- It is **indirect evidence, not a calibrated equivalence**, for the safety question. The metric is a shear-wave quality score on a phantom, not an acoustic-output measurement; it depends on displacement, which depends on the push, but through a long chain. Two



**Figure 16:** Focused B-mode of the resolution phantom, both probes, same dynamic range. Comparable wire-target resolution and penetration.

probes could differ by 10 % to 20 % in output and still score within this noise.

- If the safety numbers of §8.2 are to be applied to the MUMC probe, the honest thing is a **single hydrophone capture** on that probe at the known peak location and one transmit voltage, compared against the corresponding TU/e capture — a 15 min measurement on an aligned setup, which converts this appendix's indirect argument into a direct one.